RANDOM TRANSLATION INTERSECTIONS ON BINARY VECTOR SPACES: SHARP SPAN CLOCKS AND STABILIZER-WEIGHTED FIBRES

ANONYMOUS

Abstract. Let $V = \mathbb{F}_2^d$ and repeatedly replace a subset $A \subseteq V$ by $A \cap (A + v)$, where v is sampled uniformly from V. A length-t history acts exactly as intersection by every translate in the span of its sampled vectors. We use this sufficient statistic to give the complete rank law and a sharp worst-source absorption clock, witnessed by $V \setminus \{0\}$. For every target B, every time, and every source size, we then enumerate source–history pairs by a polynomial determined by the translation stabilizer of B. Its one-step boundary separates targets with trivial stabilizer and recovers the stabilizer dimension once the phase dimension and target size are known. The forward erosion algebra and finite-field rank law are classical inputs; the contribution assessed here is the exact conjunction with the target-dependent inverse atlas. All release claims remain under `HOLD_EXTERNAL`.

1. Process and theorem

Write $A + v = \{a + v : a \in A\}$. Starting from $A_0 \in \mathcal{P}(V)$, sample independent uniform vectors $v_1, v_2, \ldots$ and set

$$A_i = A_{i-1} \cap (A_{i-1} + v_i). \tag{1}$$

For a subspace $H \leq V$, define

$$\mathsf{E}_H(A) = \bigcap_{h \in H} (A + h), \qquad H_t = \langle v_1, \ldots, v_t \rangle.$$

Let

$$S(t, r) = \prod_{i=0}^{r-1} (2^t - 2^i), \tag{2}$$

with the empty product equal to one, and put $S(t, r) = 0$ for $r > t$. We write $\begin{bmatrix} m \\ r \end{bmatrix}_2$ for the Gaussian binomial coefficient.

For a target $B \subseteq V$, set $b = |B|$ and

$$\operatorname{Stab}(B) = \{v \in V : B + v = B\}, \qquad s = \dim \operatorname{Stab}(B).$$

The joint source-size/history polynomial is

$$F_t(B; z) = \sum_{A \subseteq V} \sum_{(v_1, \ldots, v_t) \in V^t} \mathbf{1}\{A_t = B\} z^{|A|}. \tag{3}$$

Theorem 1 (Span clock and every-target inverse atlas). *For every $d, t \geq 0$ the following statements hold.*

(1) *Every history satisfies the pointwise identity*

$$A_t = \mathsf{E}_{H_t}(A_0). \tag{4}$$

Moreover,

$$\Pr(\dim H_t = r) = \begin{bmatrix} d \\ r \end{bmatrix}_2 \frac{S(t, r)}{2^{dt}}. \tag{5}$$

(2) *The only states fixed by every possible sampled update are $\varnothing$ and V. Let $\sigma = \min\{t : H_t = V\}$. Every $A_0 \neq V$ is empty by time σ, and this bound is attained by $A_\star = V \setminus \{0\}$. Thus the sharp worst-source emptying distribution is*

$$\Pr(\sigma \leq t) = \begin{cases} 0, & t < d, \\ \displaystyle\prod_{i=0}^{d-1} (1 - 2^{i-t}), & t \geq d, \end{cases} \tag{6}$$

and

$$\mathbb{E}\sigma = \sum_{r=0}^{d-1} \frac{1}{1-2^{r-d}}. \tag{7}$$

(3) *For every target B and indeterminate z,*

$$F_t(B;z) = z^b \sum_{r=0}^{s} \begin{bmatrix} s \\ r \end{bmatrix}_2 S(t,r) \left((1+z)^{2^r} - z^{2^r}\right)^{2^{d-r}-b/2^r}. \tag{8}$$

For $r \le s$, the exponent is a nonnegative integer. Formula (8) includes $t = 0$, $d = 0$, and the empty and full targets. At one step its unweighted specialization is the boundary-safe formula

$$F_1(B;1) = \begin{cases} 1, & s = 0, \\ 1 + (2^s - 1)3^{2^{d-1}-b/2}, & s \ge 1. \end{cases} \tag{9}$$

(4) *The phase cardinality $|\mathcal{P}(V)| = 2^{2^d}$ determines d. At fixed (d,b), the value in (9) is strictly increasing over the feasible values of s, with value one occurring exactly at $s = 0$. Consequently phase size, target size, and one-step inverse mass recover the target stabilizer dimension.*

The clock in part (2) is a sharp universal bound, not an assertion that every non-full source empties exactly at σ. This distinction is important: individual sources may disappear earlier.

2. History spans and the sharp clock

The algebra of erosions by translates is standard in mathematical morphology [2, 3]; stochastic morphological sampling is also established background [4]. We record the short reduction needed for this particular process.

Lemma 2. *For subspaces $H, K \le V$, $\mathsf{E}_K(\mathsf{E}_H(A)) = \mathsf{E}_{H+K}(A)$.*

Proof. Distributing intersections over translation gives

$$\mathsf{E}_K(\mathsf{E}_H(A)) = \bigcap_{k\in K} \bigcap_{h \in H} (A + h + k) = \bigcap_{u \in H+K} (A + u).$$

Repeated factors do not affect an intersection. □

Proof of Theorem 1(1). Because $\langle v \rangle = \{0, v\}$ in characteristic two, one update is $\mathsf{E}_{\langle v \rangle}$. Lemma 2 iterated over the history proves (4).

For a fixed r-subspace H, histories spanning H are surjective linear maps $\mathbb{F}_2^t \to H$. Choosing the images dually as an ordered independent r-tuple in $\mathbb{F}_2^t$ gives (2). There are $\begin{bmatrix} d \\ r \end{bmatrix}_2$ choices of H, which proves (5). This finite-field rank enumeration is classical; we do not claim it as a standalone increment [1]. □

Proof of Theorem 1(2). If A is fixed by every update, then $A \subseteq A + v$ for every v. Cardinalities force equality, so A is invariant under all translations and hence is either $\varnothing$ or V.

When $H_t = V$, the set $\mathsf{E}_V(A)$ is V for $A = V$ and is empty otherwise. Sharpness follows from the stronger identity

$$\mathsf{E}_H(V \setminus \{0\}) = V \setminus H : \tag{10}$$

x survives exactly when $x + h \neq 0$ for every $h \in H$, equivalently when $x \notin H$. Therefore the witness empties exactly at full span.

Equation (6) is the full-rank probability in (5). At rank $r < d$, a fresh uniform vector increases the rank with probability $1 - 2^{r-d}$. The independent geometric waiting times for the successive rank increments have means $(1 - 2^{r-d})^{-1}$, proving (7). □

3. Stabilizer-weighted inverse fibres

The inverse enumeration uses information absent from the rank marginal: the specific target controls which history spans are possible.

Lemma 3 (Fixed-span fibre)**.** *Let $H \le V$ have dimension r. There exists a source A with $\mathsf{E}_H(A) = B$ if and only if $H \le \mathrm{Stab}(B)$. In that case the source-size polynomial is*

$$\sum_{A:\,\mathsf{E}_H(A)=B} z^{|A|} = z^b \left((1+z)^{2^r} - z^{2^r}\right)^{2^{d-r}-b/2^r}. \tag{11}$$

Proof. The set $\mathsf{E}_H(A)$ is the union of precisely those H-cosets wholly contained in A. Hence it is H-invariant, proving necessity. Conversely, if $H \le \mathrm{Stab}(B)$, then B is a union of $b/2^r$ full H-cosets. A source with erosion B must contain all of them. On each of the remaining $2^{d-r} - b/2^r$ cosets it may choose any proper subset, independently. One coset contributes $(1+z)^{2^r} - z^{2^r}$, which proves (11). □

Proof of Theorem 1(3). For a rank-r history, Lemma 3 permits exactly the subspaces $H \le \mathrm{Stab}(B)$. Their number is $\begin{bmatrix} s \\ r \end{bmatrix}_2$, and each is generated by $S(t,r)$ ordered histories. Summing (11) proves (8).

When $t = 0$, only $r = 0$ remains and the expression reduces to z^b, as it must. If $B = V$, the only source is V and the rank sum gives $F_t(V; z) = 2^{dt} z^{2^d}$. The empty target and $d = 0$ require no exceptional convention beyond empty products.

At $t = 1$, the zero history has rank zero and contributes one at $z = 1$. Every nonzero vector of $\mathrm{Stab}(B)$ spans a rank-one subspace, and each outside two-point coset has three proper subsets. There are $2^s - 1$ such histories. If $s \ge 1$, then B is a union of two-point orbits, so b is even and the second line of (9) follows. If $s = 0$, there is no nonzero stabilizing history and the value is one. Keeping this branch separate is essential for odd b, where the other displayed exponent would not be an integer. □

Proof of Theorem 1(4). The map $d \mapsto 2^{2^d}$ is injective. At fixed (d, b) the power of three in the second line of (9) is constant, while $2^s - 1$ is strictly increasing. Its positive contribution also separates every $s \ge 1$ from the value one at $s = 0$. □

The polynomial contains more than unweighted fibre sizes: its coefficient of z^k counts all length-t histories and k-point sources reaching the prescribed target. Thus target shape enters through $\mathrm{Stab}(B)$ even when target cardinality is held fixed.

4. Source subtraction, controls, and scope

The update's erosion interpretation and composition algebra receive zero contribution credit [2, 3]; so do the generic stochastic-morphology setting [4] and the finite-field rank distribution [1]. The bounded source audit found no primary source for the exact conjunction of the sharp witness (10), the arbitrary-target polynomial (8), and the recovery law (9). This non-hit is not a novelty, priority, or freedom-to-publish claim. Internally, the closest comparators use deterministic linear-subspace descent or random graph-cut signatures; neither transfers the affine-coset proper-subset fibre proof.

A paper-local, standard-library verifier applies every literal update and independently evaluates the span erosion over finite exhaustive boxes. It compares every target and source-size coefficient with (8), checks the rank census and full-rank boundary, tests the witness over all subspaces through dimension six, and exercises $d = 0$, $t = 0$, empty/full targets, odd-cardinality trivial stabilizers, and the repaired one-step branch. The frozen assertion total and transcript digest are recorded in the accompanying canonical file and build record. The run executes 1,712,974 integer assertions; two fresh replays were byte-identical. The canonical transcript has SHA-256 (the following two lines concatenate):

`c31ec0a098bab52241eb2765bd6fef06`
`69fdacdb4486ca69bea9dfc56fbab62b`.

Enumeration supplies counterexample pressure only; the all-parameter conclusions rest on the proofs above.

Limitations

The process uses independent uniform translations of a binary vector space and begins from an arbitrary labelled subset. We do not treat biased or dependent histories, non-elementary finite groups, noisy observations, unlabelled targets, or asymptotic distributional limits. The source search was bounded, so a direct owner under different terminology would reopen the claim boundary.

Data Availability

No external data were used. The deterministic verifier and its canonical transcript provide the complete paper-local exact control.

Ethics Statement

This mathematical study involved no human participants, animals, personal data, or field intervention.

Author Contributions

The anonymous author performed the derivation, proof, exact checks, source-boundary audit, and manuscript preparation.

Conflict of Interest

The author declares no conflict of interest.

Funding

No external funding is declared.

External Status

This anonymous internal artifact remains `HOLD_EXTERNAL`. It is not cleared for posting, submission, circulation, or author contact.

References

[1] G. V. Balakin. The distribution of the rank of random matrices over a finite field. *Theory of Probability and Its Applications*, 13(4):594–605, 1968. doi: 10.1137/1113076.

[2] H. J. A. M. Heijmans and C. Ronse. The algebraic basis of mathematical morphology I: Dilations and erosions. *Computer Vision, Graphics, and Image Processing*, 50(3):245–295, 1990. doi: 10.1016/0734-189X(90)90148-O.

[3] H. J. A. M. Heijmans and J. Serra. Convergence, continuity, and iteration in mathematical morphology. *Journal of Visual Communication and Image Representation*, 3(1):84–102, 1992. doi: 10.1016/1047-3203(92)90032-O.

[4] Krishnamoorthy Sivakumar and John Goutsias. Binary random fields, random closed sets, and morphological sampling. *IEEE Transactions on Image Processing*, 5(6):899–912, 1996. doi: 10.1109/83.503907.